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EXPERIMENT 14

INTRO TO LOG ANALYSIS

AIM:

To understand the basics of log analysis and learn how to examine system and network logs to detect suspicious activities and security incidents.

ALGORITHM:

1. Collect log files (system logs, server logs, network logs)
2. Open logs using a viewer or analysis tool
3. Identify important fields (timestamp, IP address, event type)
4. Apply filters to find specific events
5. Search for unusual or suspicious patterns
6. Correlate events to detect possible attacks
7. Extract useful information for investigation

Task 1 Introduction



Log analysis is an essential aspect of cyber security and system monitoring. At a high level, log analysis examines and interprets log event data generated by various sources (devices, applications, and systems) to monitor metrics and identify security incidents. It involves collecting, parsing, and processing log files to turn data into actionable objectives. By adopting

Answer the questions below

I'm ready to proceed!

No answer needed

✓ Correct Answer

Task 2 Log Analysis Basics

Among the various data sources collected and utilized by infrastructure systems, logs are pivotal in offering valuable insights into these systems' inner workings and interactions across the network. A log is a stream of time-sequenced messages that record occurring events. Log analysis is the process of making sense of the events captured in the logs to paint a clear picture of what has happened across the infrastructure.

What Are Logs?

Logs are recorded events or transactions within a system, device, or application. Specifically, these events can be related to application errors, system faults, audited user actions, resource uses, network connections, and more. Each log entry contains relevant details to contextualize the event, such as its timestamp (the date and time it occurred), the source (the system that generated the log), and additional information about the specific log event.

```
sample.log
Jul 28 17:45:02 10.10.0.4 FW-1: %WARNING% general: Unusual network activity detected from IP 10.10.0.15 to IP 203.0.113.25. Source Zone: I
```

In the above example, this log entry signifies an event detected by a firewall regarding unusual network activity from an internal system, indicating a potential security concern. The relevant fields to consider in this example are:

- **Server Logs:** System logs, event logs, error logs, and access logs, each offering distinct information about server operations.
- **System Logs:** Kernel activities, system errors, boot sequences, and hardware status, aiding in diagnosing system issues.
- **Network Logs:** Communication and activity within a network, capturing information about events, connections, and data transfers.
- **Database Logs:** Activities within a database system, such as queries performed, actions, and updates.
- **Web Server Logs:** Requests processed by web servers, including URLs, source IP addresses, request types, response codes, and more.

Each log type presents a unique perspective on the activities within an environment, and analyzing these logs in context to one another is crucial for effective cyber security investigation and threat detection.

Answer the questions below

I understand the basics of logs and I'm ready to proceed!

No answer needed

✓ Correct Answer

Task 3 Investigation Theory

Several methodologies, best practices, and essential techniques are employed to create a coherent timeline and conduct effective log analysis investigations.

Timeline

When conducting log analysis, creating a timeline is a fundamental aspect of understanding the sequence of events within systems, devices, and applications. At a high level, a timeline is a chronological representation of the logged events, ordered based on their occurrence. The ability to visualize a timeline is a powerful tool for contextualizing and comprehending the events that occurred over a specific period.

Answer the questions below

What's the term for a consolidated chronological view of logged events from diverse sources, often used in log analysis and digital forensics?

Super Timeline

✓ Correct Answer

Which threat intelligence indicator would `5b31f93c09ad1d065c0491b764d04933` and `763f8dbc98d105a8e82f36157e98bbe` be classified as?

File Hashes

✓ Correct Answer

Task 4 Detection Engineering

Common Log File Locations

A crucial aspect of log analysis is understanding where to locate log files generated by various applications and systems. While log file paths can vary due to system configurations, software versions, and custom settings, knowing common log file locations is essential for efficient investigation and threat detection.

- **Web Servers:**
 - **Nginx:**
 - Access Logs: `/var/log/nginx/access.log`
 - Error Logs: `/var/log/nginx/error.log`
 - **Apache:**
 - Access Logs: `/var/log/apache2/access.log`

Answer the questions below

What is the default file path to view logs regarding HTTP requests on an Nginx server?

`/var/log/nginx/access.log`

✓ Correct Answer

A log entry containing `%2E%2F%2E%2Fproc%2Fself%2Fenviron` was identified. What kind of attack might this infer?

Path Traversal

✓ Correct Answer

Task 5 Automated vs. Manual Analysis

Automated Analysis

Automated analysis involves the use of tools. For example, these often include commercial tools such as XPLG or SolarWinds Loggly. Automated analysis tools allow for processing and data analysis of logs. These tools often utilize Artificial Intelligence / Machine Learning to analyze patterns and trends. As the AI landscape evolves, we expect to see more effective automated analysis solutions.

Advantages	Disadvantages
Saves time by performing a lot of the manual work required in manual analysis	Automated analysis tools are usually commercial-only and, therefore, expensive.

Answer the questions below

A log file is processed by a tool which returns an output. What form of analysis is this?

Automated

✓ Correct Answer

An analyst opens a log file and searches for events. What form of analysis is this?

Manual

✓ Correct Answer

Task 6 Log Analysis Tools: Command Line

When analyzing collected logs, sometimes the most readily available tool we have is the command line itself. Analyzing logs through the command line provides a quick and powerful way to gain insights into system activities, troubleshoot issues, and detect security incidents, even if we don't have an SIEM system configured.

Download Task Files

Many built-in Linux commands allow us to parse and filter relevant information quickly. Viewing log files using the command line is one of the most basic yet essential tasks for conducting log analysis. Several common built-in tools are used for this purpose, offering differing functionalities to read and navigate through log files efficiently.

You can locate the `apache.log` file on the AttackBox under `/root/Rooms/introloganalysis/task6` to follow along with this task. However, it is also attached to this task and available for download.

cat

While command-line log analysis offers powerful capabilities, it might only suit some scenarios, especially when dealing with vast and complex log datasets. A dedicated log analysis solution, like the Elastic (ELK) Stack or Splunk, can be more efficient and offer additional log analysis and visualization features. However, the command line remains essential for quick and straightforward log analysis tasks.

Answer the questions below

Use `cut` on the `apache.log` file to return only the URLs. What is the flag that is returned in one of the unique entries?

`c701d43cc5a3acb9b5b04db7f1be94f6`

✓ Correct Answer



In the `apache.log` file, how many total HTTP 200 responses were logged?

`52`

✓ Correct Answer



In the `apache.log` file, which IP address generated the most traffic?

`145.76.33.201`

✓ Correct Answer



What is the complete timestamp of the entry where `110.122.65.76` accessed `/login.php`?

`31/Jul/2023:12:34:40 +0000`

✓ Correct Answer



Task 7 Log Analysis Tools: Regular Expressions



Answer the questions below

How would you modify the original `grep` pattern above to match blog posts with an ID between 20-29?

`post=2[0-9]`

✓ Correct Answer



What is the name of the filter plugin used in Logstash to parse unstructured log data?

`Grok`

✓ Correct Answer

Task 8 Log Analysis Tools: CyberChef



CyberChef is a powerful tool in an analyst's toolkit. Created by GCHQ, CyberChef has long been regarded as the "Cyber Swiss Army Knife." The application boasts over 300 operations, which combine to make a recipe that makes handling data a piece of cake. Some key features include:

[Download Task Files](#)

- Encoding and decoding data
- Encryption and hashing algorithms
- Data analysis, such as parsing log files and extracting data
- And many more!

This task is going to demonstrate how CyberChef can be used to parse a log file alongside the use of recipes for analysis. Before we begin, let's become familiar with the CyberChef interface. First, let's launch CyberChef in our browser by visiting [CyberChef](#). Note, if you are a free user on the AttackBox, a local copy of CyberChef is installed and can be accessed by clicking the "Offline CyberChef" bookmark in Firefox.

Answer the questions below

Locate the "loganalysis.zip" file under `/root/Rooms/introloganalysis/task8` and extract the contents.

No answer needed

✓ Correct Answer

Upload the log file named "access.log" to CyberChef. Use regex to list all of the IP addresses. What is the full IP address beginning in 212?

212.14.17.145

✓ Correct Answer

Using the same log file from Question #2, a request was made that is encoded in base64. What is the decoded value?

THM{CYBERCHEF_WIZARD}

✓ Correct Answer

Using CyberChef, decode the file named "encodedflag.txt" and use regex to extract by MAC address. What is the extracted value?

08-2E-9A-4B-7F-61

✓ Correct Answer

Task 9 ✓ Log Analysis Tools: Yara and Sigma

Sigma

Sigma is a highly flexible open-source tool that describes log events in a structured format. Sigma can be used to find entries in log files using pattern matching. Sigma is used to:

This YARA rule can be expanded to look for:

- Multiple IP addresses
- IP Addresses based on a range (for example, an ASN or a subnet)
- IP addresses in HEX
- If an IP address lists more than a certain amount (i.e., alert if an IP address is found five times)
- And combined with other rules. For example, if an IP address visits a specific page or does a certain action

If you want to learn more about Yara, check out the [Yara](#) room on TryHackMe.

Answer the questions below

What languages does Sigma use?

YAML

✓ Correct Answer

What keyword is used to denote the "title" of a Sigma rule?

title

✓ Correct Answer

What keyword is used to denote the "name" of a rule in YARA?

rule

✓ Correct Answer

Task 10 ✓ Conclusion

In this room, we covered the basic methodology behind adopting an effective log analysis strategy. We explored the importance of log data collection, common attack patterns, and useful tools for the investigation and response processes.

Next Steps

To expand your SIEM and centralized logging solution capabilities, visit the [Advanced Splunk](#) and [Advanced ELK](#) modules.

Answer the questions below

Click and continue learning!

No answer needed

✓ Correct Answer

RESULT:

Successfully analyzed log data to identify unusual activities and gained basic knowledge of detecting security incidents through log analysis.